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Enriching Structured Dataset via Reasoning Embeddings with Few-Shot Learning for Enhancing Complaint Classification

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Abstract

Public complaint classification is essential for effective municipal governance, yet remains challenging due to heterogeneous structured data, severe class imbalance, and limited ability of structured features to capture underlying representation. Unlike the existing studies that relied solely on the original features for end-to-end prediction, the contribution of this study lied in transforming large language models (LLMs) generated reasoning under few-shot learning into an explicit feature engineering step, where reasoning outputs are integrated with structured data and embedded to enrich representations. Extensive experiments were conducted on a real-world municipal complaint dataset containing over 48103 instances across 160 complaint subcategories. The best-performing configuration which combined Qwen3-14B, Paraphrase-Multilingual-Mpnet-Base-V2 embeddings, and Logistic Regression, achieved an average score of 44.8%, with an accuracy of 54%, an F1-score of 23.9%, and an Area Under the Curve of 92%. Moreover, error analysis revealed low class-wise error rates alongside reduced false positive and false negative rates, indicating balanced control over both false alarms and missed detections. The Shapiro-Wilk test rejected normality (statistic ≥ 0.873 , $p < 0.05$), while the Friedman test indicated significant differences across models (statistic ≥ 182.855 , $p < 0.05$), further supported by non-overlapping confidence intervals with F1-score gains from 7.3% - 7.8% to 12% - 12.7%. Ablation studies highlight the critical role of each component of proposed framework, improving accuracy from 34.4% to 37.5% and F1-score from 12.5% to

14.1%. Overall, these findings demonstrated that LLM-generated reasoning effectively enhanced structured representations and predictive performance of existing Machine Learning classifiers in public complaint classification.

Keywords: Large Language Model, Reasoning, Embeddings, Complaint Classification, Few-Shot Learning, Machine Learning

1 Introduction

Public complaints constitute a primary communication channel through which citizens convey dissatisfaction, report service deficiencies, or request interventions from governmental and municipal authorities [1–6]. These submissions commonly span a wide range of domains, including infrastructure maintenance, public safety, environmental management, healthcare services, and administrative operations [6–9]. Effective categorization of such complaints is essential for enabling timely prioritization, efficient resource allocation, and coordinated responses by responsible agencies [5, 7, 10]. Despite their importance, the automatic analysis of public complaints remains challenging. Complaints are often articulated in heterogeneous textual forms, varying in length, structure, and linguistic style, and are frequently submitted in multiple languages or dialects [3]. As a result, many public-sector organizations continue to rely on manual or semi-automated workflows for complaint handling. These approaches are labor-intensive, costly, and prone to delays, particularly in settings where human resources are limited [7, 10, 11]. Delayed or inaccurate responses not only hinder service delivery but may also amplify public dissatisfaction and erode trust in governmental institutions. Moreover, manual categorization is susceptible to subjectivity and human error, which can lead to inconsistent labeling and the potential neglect of critical issues [2, 3]. Collectively, these limitations highlight the pressing need for intelligent, scalable, and reliable methods for automated complaint classification.

In response to these challenges, prior research has explored a variety of computational approaches for complaint and text classification, ranging from Machine Learning (ML) models to more recent Deep Learning (DL) architectures. While these methods have demonstrated varying degrees of success, their performance often depends heavily on extensive features and carefully engineered features, which are not always readily available in real-world public-sector scenarios. More recently, large language models (LLMs) have emerged as an effective alternative, owing to their strong capability in capturing semantic context and linguistic nuances across diverse domains [2, 12–24]. LLMs have shown notable effectiveness in understanding short, noisy, and informal texts which are the characteristics that are commonly observed in public complaints. Existing studies have predominantly leveraged LLMs within few-shot or prompting-based paradigms, treating them as end-to-end predictors that directly generate complaint category labels [12, 23–32].

Despite these advances, such approaches largely overlook an important aspect of LLM capabilities. Specifically, the intermediate reasoning responses generated during inference are typically discarded and not utilized beyond the prediction stage. As a

result, LLMs are primarily used as black-box classifiers, and their potential to provide rich, structured semantic explanations remains underexploited. This limitation highlights a critical research gap. The role of LLM-generated reasoning as a form of semantic feature engineering has not been systematically investigated in the context of public complaint classification. While related domains such as healthcare and education have demonstrated that incorporating contextual LLM outputs such as embedding values can enrich structured datasets and improve downstream learning, this paradigm has yet to be effectively adapted for complaint analytics.

To address this gap, this work proposes a reasoning-enhanced representation learning framework for structured complaint classification. The core idea is to move beyond treating LLMs solely as prediction tools, and instead leverage their reasoning capability as a source of auxiliary semantic features. Specifically, under a few-shot learning setting, the proposed method extracts reasoning responses generated by LLM, integrates with existing structured complaint features and encodes them into dense embeddings to enrich the overall representation space. By incorporating both structured attributes and semantically meaningful reasoning signals, the framework enables improved feature expressiveness and enhances the model's ability to discriminate between complaint categories.

The novelty of this work lies in the systematic integration of LLM-generated reasoning into a structured learning pipeline, transforming implicit reasoning into explicit and learnable representations. Unlike existing LLM-based classification approaches, which focus solely on output labels, the proposed framework utilizes intermediate reasoning as an additional information source for downstream models. Extensive empirical evaluations on real-world complaint datasets demonstrate that the proposed approach consistently outperforms competitive baselines. These findings highlight the effectiveness of reasoning-enhanced representations and establish a new direction for leveraging LLMs in structured classification tasks.

The remainder of this paper is organized as follows. Section 2 reviewed the existing studies on public complaint classification with particular emphasis on approaches leveraging LLMs and critically analyzed their limitations in capturing representations from structured complaint data. This review highlights key challenges related to feature representation that motivate the proposed reasoning embeddings framework. Section 3 described the proposed framework that details the few-shot learning used to elicit LLM-generated reasoning, the construction of reasoning embeddings, and their integration with structured complaint features for complaint classification. Section 4 presented the experimental results and discussion, providing a comprehensive analysis of classification performance and contribution of reasoning embeddings relative to competitive baselines. Finally, Section 5 concluded the paper by summarizing the main findings, discussing practical implications for public-sector complaint management, and outlining potential directions for future research.

2 Literature Review

Existing studies have increasingly explored the use of LLMs to transform structured tabular features into semantic embeddings. In healthcare, the researchers demonstrated

that the LLM-generated embeddings achieve performance comparable to models trained directly on raw tabular features for medical machine learning tasks [19]. In related study, the researchers reformulated tabular datasets into two-dimensional embedding representations and employed Convolutional Neural Networks (CNNs) for downstream prediction, showing that domain-specific LLM embeddings consistently outperform one-hot encodings [18]. Notably, this approach applies CNNs to tabular data rather than their common use in computer vision or pattern recognition tasks, where inputs are typically two-dimensional images in recent work [33–36]. Beyond strictly structured dataset, Sakai et al. [20] reported that Generative Pre-trained Transformer (GPT) models greatly outperformed existing ML models on multilabel classification of patient comments, achieving F1-score approximately 76%. These findings collectively highlight the representational power of LLM embeddings across both structured and semi-structured learning scenarios. The effectiveness of LLM-based representations has also been observed beyond healthcare. Reasoning-capable LLMs incorporating chain-of-thought (CoT) prompting have achieved superior accuracy in consumer complaint classification, marking a significant advancement in automated complaint handling systems [2]. In education analytics, Bidirectional Encoder Representations from Transformers (BERT) cderived embeddings combined with Boruta feature selection achieved classification accuracies of approximately 88% for predicting student outcomes [21]. Across these studies, LLM-generated embeddings often rank among the most predictive features [21, 22]. To further enrich the structured data, the researchers have appended contextual embeddings generated by LLMs such as GPT-2 to tabular inputs, reporting substantial performance gains when paired with models such as Extreme Gradient Boosting (XGBoost) and Categorical Boosting (CatBoost). These results consistently indicate that LLM-based features enhance predictive performance, particularly in domains characterized by complex semantics or limited labeled datasets.

By contrast, the application of LLM-derived embeddings to public complaint classification remains relatively limited. Existing studies have largely focused on embedding the raw textual content of complaints rather than transforming structured complaint attributes into semantic representations via LLMs. More broadly, current complaint classification systems primarily rely on natural language processing (NLP) pipelines or fine-tuned LLMs operating directly on complaint text [12–17]. Nevertheless, evidence from adjacent domains suggests strong potential for extending LLM-based embedding techniques to public complaints. For instance, consumer complaint classification in Roumeliotis et al. [2] and the transport-related complaint analysis by Rakhimzhanov et al. [12] demonstrate that LLM-based representations can substantially improve classification accuracy in linguistically rich and high-stakes contexts. Moreover, the effectiveness of LLM embeddings in healthcare where models effectively encode numerical vitals and categorical symptom codes suggests that structured complaint features could likewise be captured if reformulated as language-based inputs to LLMs. Overall, while direct studies on LLM-generated embeddings for public complaint data are sparse, accumulating evidence from related domains indicates that LLM-encoded representations of structured inputs are both effective and promising [2, 18–22].

Parallel to advances in embedding-based representations, recent research has examined the use of few-shot learning with LLMs for complaint classification. Rakhimzhanov et al. [12] directly compared few-shot LLM inference with multilingual embedding-based classifiers on a public transport complaint dataset, concluding that embedding-based approaches offer a more practical alternative when computational efficiency and scalability are critical, despite few-shot learning yielding reasonable performance. Similarly, other studies compared zero-shot inference against few-shot fine-tuning on small in-domain civic grievance datasets and found that zero-shot inference consistently outperformed few-shot configurations with 8, 32, and 128 examples [23]. These findings suggest that in-context adaptation may degrade generalization when labeled data are insufficient, cautioning against over-reliance on few-shot fine-tuning in low-resource complaint classification settings. Additional work has applied LLM prompting strategies to related complaint classification tasks. For example, GPT-based models have been used to classify healthcare complaints according to the Healthcare Complaint Analysis Tool (HCAT) taxonomy by incorporating category definitions and keywords into the prompt [24]. Under zero-shot settings, this approach achieved classification accuracies ranging from 58.4% to 95.5%, despite not employing explicit few-shot examples or step-by-step reasoning [24]. While effective, these results also reveal a limitation of direct prompting approaches: linguistic variation, human phrasing, and implicit bias in complaint text may influence model predictions unless supplemented by examples or structured reasoning mechanisms [24, 27–31].

A broader and emerging trend is the use of reasoning-based LLMs for complaint classification. Prior studies evaluated multiple reasoning-capable models, including GPT variants, on the Consumer Financial Protection Bureau (CFPB) complaints dataset [2]. These models explicitly decompose the classification task into intermediate reasoning steps, which have been shown to improve performance when complaint categories are overlapping or implicitly defined [2, 24]. Furthermore, combining Knowledge Graphs (KGs) with CoT reasoning achieved the highest F1-scores for most complaint categories, although few-shot learning performed better in certain ambiguous categories due to their broad definitions and susceptibility to noise [37]. These findings underscore the importance of clearer semantic distinctions among complaint types and demonstrate that reasoning-enhanced LLMs can mitigate misclassification in complex category structures [2, 24, 37].

Despite these advances, existing research largely treats few-shot learning and reasoning as end-to-end prediction mechanisms, where LLMs directly output complaint categories [12, 23–32]. The intermediate reasoning responses generated during inference are typically discarded and not incorporated into downstream learning pipelines. Consequently, the potential of LLM-generated reasoning as a semantic feature engineering mechanism remains largely unexplored. While related domains such as healthcare and education have shown that embedding explanatory or contextual LLM outputs can enrich structured datasets, this paradigm has not yet been systematically applied to public complaint classification.

In summary, prior research reveals three key trends: (i) LLMs are effective in generating embeddings that enrich structured and tabular data [21]; (ii) few-shot learning

and reasoning-enhanced LLMs improve complaint classification under limited supervision [2, 12, 23, 24]; and (iii) embedding-based approaches often provide superior efficiency compared to direct LLM inference [2, 12–22, 38]. However, there is limited work that integrates these advances into a unified framework for public complaint classification. Existing studies do not sufficiently explore enriching tabular complaint datasets by appending LLM-generated reasoning responses produced via few-shot learning to original structured features and encoding them as embeddings. This gap motivates the present study, which proposes a framework for enriching structured complaint datasets through LLM-generated reasoning embeddings, bridging reasoning-capable LLMs and existing ML models for accurate, scalable, and efficient complaint classification.

3 Method

This section presents the research methodology adopted to investigate the effectiveness of enriching structured complaint datasets with reasoning-embeddings representations for complaint classification. The methodology is designed to systematically integrate reasoning-capable LLMs with existing embedding models and ML classifiers, enabling a rigorous evaluation of how intermediate reasoning can enhance semantic representation learning. Given the highly imbalanced nature of public complaint data and the coexistence of structured categorical features with unstructured textual fields, the proposed methodology emphasizes generality, scalability, and fairness across class-frequency settings. To this end, the experimental design combines real-world municipal complaint data, few-shot learning strategies, multiple reasoning-capable LLMs, diverse embedding models, and a broad range of classifiers to ensure that observed performance gains can be attributed to the proposed reasoning-enriched framework rather than to model-specific biases or dataset characteristics.

Figure 1 illustrates the high-level architecture of the proposed framework which consists of three tightly coupled stages such as data collection, reasoning embedding construction, and complaint subcategories classification. First, each complaint instance is represented using its original structured features, including administrative metadata and textual descriptions. These attributes are reformulated into a unified textual representation to enable compatibility with language models. Second, a reasoning-capable LLM is conditioned using a few-shot learning strategy and is prompted to generate an explicit reasoning response for each complaint instance. This reasoning captures latent intent, contextual cues, and semantic relationships that are not explicitly encoded in the original structured features. The generated reasoning is then concatenated with the original complaint features and encoded into a dense vector using a sentence embedding model to produce a reasoning-enriched representation. Finally, these embeddings are used as inputs to existing ML classifiers to predict complaint subcategories, allowing the effectiveness of the enriched representations to be evaluated independently of end-to-end LLM inference.

In summary, the proposed methodology establishes a unified and modular framework that bridges reasoning-capable LLMs and existing ML pipelines. By explicitly decoupling reasoning generation, embedding construction, and classification, the

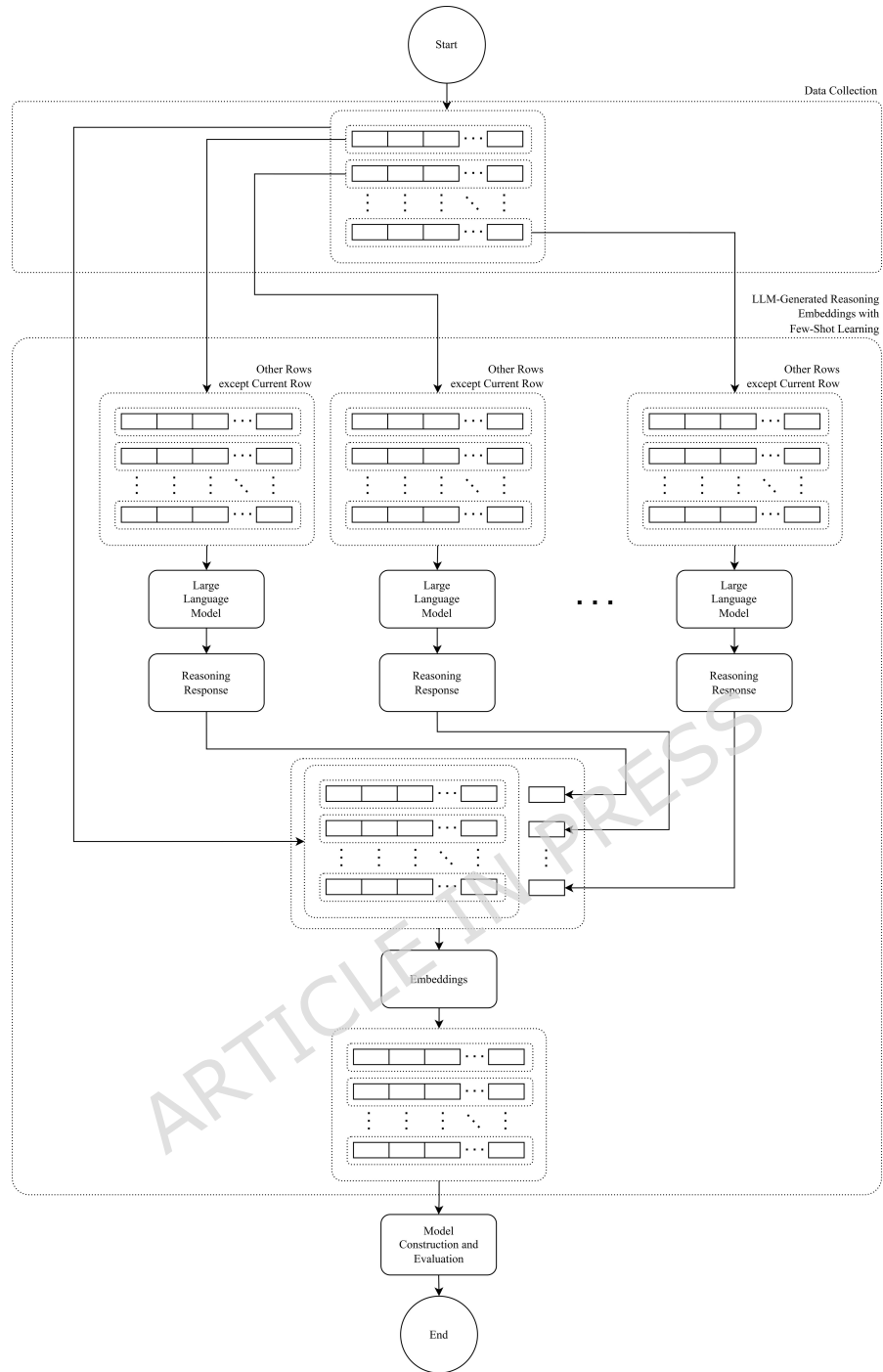


Fig. 1 High-Level Framework

framework enables detailed analysis of each component’s contribution while maintaining flexibility in model selection and deployment. This design not only facilitates a fair comparison against non-reasoning baselines but also ensures that the approach remains practical for real-world complaint management systems, where interpretability, scalability, and effectiveness across evolving complaint distributions are critical considerations.

3.1 Data Collection

Table 1 Feature Descriptions for Complaint Dataset

Feature	Type	Description
Date	Date	The timestamp when the complaint was filed
Title	Categorical	A concise summary of the core issue of the complaint
Source	Categorical	The medium or platform through which the complaint was submitted
District	Categorical	The administrative district where the complaint originated
State Legislative Assembly	Categorical	The legislative body or governance unit associated with the jurisdiction of the complaint
Details	Categorical	A detailed textual field providing in-depth context and description of the complaint
Address	Categorical	The physical address related to the complaint
Department	Categorical	The governmental or organizational department responsible for addressing the complaint
Main Category	Categorical	A broad classification of the complaint
Category	Categorical	A more specific subdivision under the Main Category

The empirical evaluation in this study is conducted on a real-world complaint dataset collected from one of the municipal service providers in Seberang Perai, Malaysia. The dataset comprises a total of 48103 individual complaints recorded over an 18-month period, spanning from 1 January 2024 to 30 June 2025. Each complaint instance is represented using ten structured attributes, as summarized in Table 1, capturing both administrative metadata and rich textual information. Specifically, the dataset includes temporal information (Date), concise problem summaries (Title), submission channels (Source), and multiple location-related attributes such as District, State Legislative Assembly, and Address. In addition, organizational and semantic dimensions are represented through the Department, Main Category, and Category fields, while the Details attribute provides a free-text description offering in-depth contextual information about each complaint. This combination of structured categorical features and unstructured textual content makes the dataset particularly suitable for investigating representation learning and reasoning-based enrichment techniques.

In this work, the complaint subcategory is used as the target variable for classification. The dataset contains 180 distinct subcategories, reflecting a wide spectrum of

Table 2 Sample Complaints Subcategories and Their Frequency Between 1 January 2024 and 30 June 2025

Subcategory	Between 1 January 2024 and 31 May 2025	Between 1 June 2025 and 30 June 2025
other complaints related to tree	4204	229
vehicle obstruction	4029	256
stray / roaming animals	3156	179
dirty / clogged drain	2106	108
streetlight / pole damage	1805	36
public nuisance	1435	97
traffic light / pole damage	1219	78
bushes/weeds	1151	51
delay in garbage collection (landfill)	1125	58
other complaints (directorate department)	1109	97
hawker license	1	0
earthworks (management services)	1	0
tree cutting work not fully completed	1	0
traffic signs	1	0
job application	1	0
building & land valuation	1	0
building department	1	0
sp arena seberang jaya	1	0
seberang jaya swimming complex	1	0
other complaints related to swiftlet birds	1	0

public service issues ranging from infrastructure damage and environmental concerns to licensing and administrative matters. To ensure a validation strategy that reflects real-world conditions, the dataset is partitioned using a temporal split, where records from 1 January 2024 to 31 May 2025 are used for training, while data from 1 June 2025 to 30 June 2025 are reserved as the testing set, which serves both as the validation and final evaluation dataset to assess the performance of the proposed framework. Table 2 reports the frequency distribution of selected sample subcategories across two-time intervals. As shown, the distribution is highly imbalanced, with a small number of dominant subcategories such as *other complaints related to tree*, *vehicle obstruction*, and *stray / roaming animals* accounting for a substantial portion of the total volume, while many subcategories occur only once or a few times throughout the entire collection period. This long-tailed distribution poses significant challenges for existing supervised learning approaches, particularly for rare classes with extremely limited labeled instances.

3.2 LLM-Generated Reasoning Embeddings with Few-Shot Learning

To enrich the structured complaint data with semantically reasoning-aware representations, an LLM-generated reasoning embedding framework with few-shot learning

is proposed as formalized in Algorithm 1. The core idea is to condition a reasoning-capable LLM on a small, class-balanced support set and to use the model’s generated reasoning as an intermediate semantic representation for embedding construction. This process transforms each structured complaint instance into a dense representation that encodes both its original features and contextualized reasoning derived from similar historical cases. Let $X = \{x_1, x_2, \dots, x_N\}$ denote the structured feature matrix of N complaint instances, $y = \{y_1, y_2, \dots, y_N\}$ represent the corresponding target labels, L be a reasoning-capable LLM, T be a text embedding model, and K be the number of shots per class used for few-shot conditioning; the output of the algorithm is the set of reasoning-enriched embeddings $E = \{e_1, e_2, \dots, e_N\}$, where each e_i is a dense vector representation associated with instance x_i .

Algorithm 1 LLM-Generated Reasoning Embeddings with Few-Shot Learning

Input

X Structured feature matrix $\{x_1, x_2, \dots, x_N\}$
 y Target labels $\{y_1, y_2, \dots, y_N\}$
 L Reasoning-capable LLM
 T Text embedding model
 K Number of shots per class

Output

E Reasoning-enriched embeddings $\{e_1, e_2, \dots, e_N\}$

```

1: function REASONINGEMBEDDINGS( $X, y, L, T, K$ )
2:    $X_c \leftarrow$  copy of  $X$ 
3:    $E \leftarrow$  empty list
4:   for each  $x_i \in X_c$  do
5:      $D_{-i} \leftarrow \{(x_j, y_j) | j \neq i\}$ 
6:      $M \leftarrow$  empty list
7:      $u \leftarrow$  unique classes of  $y$ 
8:     for each  $c \in u$  do
9:        $\{(x_j, y_j)\}_{j=1}^K \subset \{(x_j, y_j) \in D_{-i} | y_j = c\}$ 
10:      Append the selected  $K$  instances to  $M$ 
11:    end for
12:     $l_i \leftarrow$  condition  $L$  on samples  $M$ 
13:     $r_i \leftarrow$  extract the reasoning response of  $l_i$  for  $x_i$ 
14:     $t_i \leftarrow$  convert  $x_i$  with  $r_i$  to a comma-separated text string
15:     $e_i \leftarrow$  generate embeddings of  $t_i$  using  $T$ 
16:    Append  $e_i$  to the list  $E$ 
17:  end for
18:  return  $E$ 
19: end function

```

Algorithm 1 begins by creating a working copy of the structured feature matrix X (Algorithm 1, Line 2) to ensure that the original dataset remains unchanged throughout the enrichment process. An empty list E is then initialized to store the resulting

reasoning-enriched embeddings for all complaint instances (Algorithm 1, Line 3). The algorithm iterates over each instance $x_i \in X_c$ (Algorithm 1, Line 4), treating it as the query instance for which a reasoning-aware embedding is to be generated. For each x_i , a leave-one-out strategy is applied by constructing a reduced dataset D_{-i} that excludes the current instance (Algorithm 1, Line 5). This design choice ensures that no label or content leakage occurs during reasoning generation, while simultaneously maximizing the number of available training samples **for the subsequent process**. An empty memory buffer M is then initialized to store few-shot examples (Algorithm 1, Line 6), and the set of unique class labels u is extracted from the target label vector y . This step establishes the foundation for balanced few-shot conditioning across all complaint subcategories.

The algorithm then iterates over each class $c \in u$ (Algorithm 1, Line 8). For every class, K instances are sampled from D_{-i} such that their labels match the current class c (Algorithm 1, Line 9). These instances collectively form the few-shot support examples and are appended to the memory buffer M (Algorithm 1, Line 10). By repeating this process for all classes, the algorithm constructs a class-aware few-shot context that exposes the LLM to representative patterns from each complaint subcategory. In this study, the leave-one-out formulation **is adopted for few-shot learning, where $K = N - 1$ to ensure the** effective use of all available data except the query instance itself, which is particularly important for rare classes with limited samples.

Once the support set M is assembled, **each example is concatenated into a comma-separated textual string and collectively served as the prompt input to the LLM L under a few-shot learning setting, resulting a LLM l_i that conditioned on samples M (Algorithm 1, Line 12)**. This conditioning step primes the model with task-specific and class-discriminative information, enabling it to perform contextual reasoning over the query instance. **The query instance x_i is likewise concatenated into a comma-separated textual string and provided as input to l_i , which then generates a reasoning response r_i (Algorithm 1, Line 13) that typically captures latent relationships, inferred intent, and semantic cues that are not explicitly encoded in the structured features.** To integrate this reasoning representation with the original data, the structured attributes of x_i are concatenated with the generated reasoning r_i into a single comma-separated textual representation t_i (Algorithm 1, Line 14). This transformation bridges structured and unstructured modalities, allowing downstream embedding models to process both feature-level information and high-level semantic reasoning in a unified manner. The text embedding model T is then applied to t_i to generate the final reasoning-enriched embedding e_i (Algorithm 1, Line 15), which is subsequently appended to the embedding list E (Algorithm 1, Line 16).

After all instances have been processed, the algorithm returns the complete set of reasoning-enriched embeddings E (Algorithm 1, Line 18). These embeddings serve as enhanced representations for subsequent complaint classification tasks. **From a computational perspective, the proposed leave-one-out few-shot strategy incurs a time complexity of $\mathcal{O}(N^2)$ in the worst case, as for each query instance x_i , the algorithm constructs a support set using up to $N - 1$ remaining samples. The dominant cost arises from repeatedly conditioning the LLM and generating reasoning responses for each instance, leading to $\mathcal{O}(N \cdot C_L)$ complexity, where C_L denotes the cost of a single**

LLM inference with a prompt of size proportional to N . Additionally, the embedding generation step contributes $\mathcal{O}(N \cdot C_T)$, where C_T is the cost of the text embedding model. By explicitly incorporating LLM-generated reasoning under a few-shot learning paradigm, the proposed approach is able to mitigate class imbalance, improve semantic separability among subcategories, and provide richer representations than conventional feature-only or text-only embeddings. This makes the framework particularly effective for real-world complaint datasets characterized by long-tailed distributions and limited labeled examples per class.

3.3 Model Construction and Evaluation

This study evaluates the effectiveness of the proposed reasoning-enriched representation framework through a comprehensive model construction and benchmarking pipeline. Multiple reasoning-capable LLMs are employed to generate intermediate reasoning responses, including DeepSeek-R1-14B, Cogito-14B, Qwen3-14B, GPT-OSS-20B, and Magistral-24B. These models are selected to reflect diverse architectural designs and parameter scales, enabling an effective assessment of how different reasoning capabilities influence downstream embedding quality and classification performance. Each LLM is conditioned using the few-shot strategy described in the previous subsection and is applied consistently across all experiments to ensure fair comparison.

Table 3 Embedding Vector Dimension

Embedding Model	Embedding Vector Dimension
All-MiniLM-L12-V2	384
All-Mpnet-Base-V2	768
All-DistilRoberta-V1	768
Paraphrase-Multilingual-MiniLM-L12-V2	384
Paraphrase-Multilingual-Mpnet-Base-V2	768

Following reasoning generation, complaint instances are transformed into dense vector representations using a set of widely adopted sentence embedding models with varying embedding dimensionalities, as summarized in Table 3. Specifically, All-MiniLM-L12-V2 (MiniLM) and Paraphrase-Multilingual-MiniLM-L12-V2 (Pr-M-LM) generate compact 384-dimensional embeddings, while All-Mpnet-Base-V2 (Mpnet), All-DistilRoberta-V1 (DRoberta), and Paraphrase-Multilingual-Mpnet-Base-V2 (Pr-M-Mp) produce higher-capacity 768-dimensional representations. This diversity investigates the interaction between embedding dimensionality, semantic richness, and classification effectiveness when combined with LLM-generated reasoning. For the downstream classification task, a broad spectrum of ML classifiers is employed, covering both linear and non-linear paradigms. These include Logistic Regression (LR), Gaussian Naïve Bayes (GNB), Random Forest (RF), Adaptive Boosting (AdaBoost), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LGBM),

and Categorical Boosting (CatBoost). All classifiers are trained on the generated embedding representations, enabling a systematic evaluation of how different learning algorithms exploit reasoning-enriched features under highly imbalanced, multi-class settings in predicting the complaint subcategories.

As a benchmark for comparison, the proposed reasoning-enriched framework is evaluated against a baseline representation constructed by converting each complaint’s original structured features, as listed in Table 1, into a single comma-separated textual string without any LLM-generated reasoning. These baseline textual representations are embedded using the same set of embedding models and classified using identical ML algorithms. This controlled comparison isolates the contribution of LLM-generated reasoning and few-shot conditioning, ensuring that any observed improvements can be attributed to semantic enrichment rather than differences in model capacity or training procedures. Model performance is assessed using a comprehensive set of evaluation metrics, including accuracy, precision, recall, F1-score, and the Area Under the Curve (AUC).

Building upon these metrics, the evaluation is organized to progressively examine performance at increasing levels of granularity. First, an overall comparative analysis is conducted to evaluate average classification performance across different reasoning-capable LLMs, embedding models, and evaluation metrics, thereby establishing the general effectiveness of reasoning enrichment. This is followed by a frequency-aware analysis, which investigates how the proposed framework performs across minority and majority complaint subcategories, highlighting its effectiveness under varying degrees of data scarcity and its ability to mitigate long-tailed distribution effects. Subsequently, a ranking-based comparative analysis of end-to-end model configurations is performed, offering an operational perspective by identifying the most effective and least effective pipelines when reasoning strategies, embeddings, and classifiers are jointly considered.

Building on this comparative foundation, an error analysis is conducted to provide a more granular examination of class-specific misclassification patterns with true positive (TP), true negative (TN), false positive (FP) and false negative (FN). This analysis incorporates false positive rate (FPR), which reflects the proportion of non-relevant cases that are incorrectly identified as positive (Equation 1), and false negative rate (FNR), which measures the proportion of actual positive cases that are incorrectly missed by the model (Equation 2). In practical terms, a low FNR indicates that the model effectively captures relevant complaint categories, particularly minority classes, while a low FPR ensures that it avoids incorrectly assigning complaints to inappropriate categories. In addition, the overall Error represents the proportion of total misclassifications (Equation 3), capturing both over-prediction (false positives) and under-prediction (false negatives). Together, these measures provide a comprehensive understanding of prediction quality beyond aggregate metrics, enabling the identification of systematic failure modes across challenging categories.

$$FPR = \frac{FP}{FP + TN} \quad (1)$$

$$FNR = \frac{FN}{FN + TP} \quad (2)$$

$$Error = \frac{FP + FN}{TP + TN + FP + FN} \quad (3)$$

To ensure that the observed performance differences are statistically sound, statistical significance testing is subsequently performed. The Shapiro-Wilk test is first applied to assess normality under the null hypothesis H_0 : the data are normally distributed, against the alternative hypothesis H_1 : the data are not normally distributed. Upon rejection of H_0 when p-value $p < 0.05$, indicating non-normality, parametric assumptions are violated; therefore, an ANOVA test is considered when approximate normality is assumed under large-sample conditions, while the Friedman test is adopted as a non-parametric alternative for comparing multiple related models. In both tests, the null hypothesis H_0 states that all models perform equivalently, whereas the alternative hypothesis H_1 indicates that at least one model performs significantly differently. Complementing these tests, confidence interval analysis is incorporated to quantify the reliability and variability of each evaluation metric, providing further evidence of performance consistency.

Finally, an ablation study is conducted to isolate and quantify the individual contributions of few-shot learning and reasoning components within the proposed framework. By systematically removing or modifying these components while keeping other factors constant, this analysis clarifies their respective roles and interactions in driving performance improvements. Collectively, this multi-level evaluation process ensures that the effectiveness of the proposed framework is not only demonstrated through empirical results but also rigorously validated and thoroughly interpreted, offering comprehensive insights into its strengths, limitations, and practical applicability in structured complaint classification systems.

4 Results and Discussions

This section presents a comprehensive analysis of the experimental results and discusses the effectiveness of the proposed reasoning-enriched framework for public complaint classification. The evaluation is designed to rigorously assess whether incorporating LLM-generated reasoning embeddings under a few-shot learning leads to consistent and meaningful performance improvements over existing embedding-based baselines. Given the extreme class imbalance, large number of complaint subcategories, and heterogeneous nature of the structured features, the results are analyzed from multiple complementary perspectives to provide a deeper understanding of model behavior, generalization ability, and practical applicability.

4.1 Comparative Analysis of Average Classification Performance Across Reasoning–Embedding Configurations

Table 4 presents a comprehensive comparison of the average classification performance achieved with and without the proposed reasoning-enriched framework across multiple embedding models and evaluation metrics. The results clearly demonstrate that incorporating LLM-generated reasoning consistently improves performance over the non-reasoning baseline for nearly all embedding configurations. In the absence

Table 4 Average Performance (mean $\pm$ standard deviation) across Reasoning-Capable LLMs and Embedding Models, Compared against a Non-Reasoning Baseline

Reasoning	Embedding	Accuracy	Precision	Recall	F1-score	AUC
No	MiniLM	0.307 $\pm$ 0.154	0.127 $\pm$ 0.085	0.111 $\pm$ 0.066	0.093 $\pm$ 0.062	0.720 $\pm$ 0.139
	Mpnet	0.281 $\pm$ 0.150	0.118 $\pm$ 0.083	0.094 $\pm$ 0.060	0.082 $\pm$ 0.060	0.698 $\pm$ 0.122
	DRoberta	0.263 $\pm$ 0.145	0.098 $\pm$ 0.064	0.082 $\pm$ 0.050	0.070 $\pm$ 0.048	0.686 $\pm$ 0.122
	Pr-M-LM	0.326 $\pm$ 0.157	0.140 $\pm$ 0.090	0.132 $\pm$ 0.079	0.112 $\pm$ 0.073	0.759 $\pm$ 0.152
DeepSeek-R1-14B	Pr-M-Mp	0.344 $\pm$ 0.171	0.146 $\pm$ 0.098	0.146 $\pm$ 0.091	0.125 $\pm$ 0.083	0.774 $\pm$ 0.155
	MiniLM	0.317 $\pm$ 0.152	0.136 $\pm$ 0.087	0.119 $\pm$ 0.071	0.103 $\pm$ 0.067	0.732 $\pm$ 0.145
	Mpnet	0.357 $\pm$ 0.161	0.149 $\pm$ 0.097	0.154 $\pm$ 0.091	0.126 $\pm$ 0.081	0.769 $\pm$ 0.152
	DRoberta	0.351 $\pm$ 0.119	0.139 $\pm$ 0.083	0.131 $\pm$ 0.076	0.111 $\pm$ 0.068	0.747 $\pm$ 0.145
Cogito-14B	Pr-M-LM	0.338 $\pm$ 0.147	0.135 $\pm$ 0.097	0.139 $\pm$ 0.089	0.116 $\pm$ 0.081	0.770 $\pm$ 0.158
	Pr-M-Mp	0.371 $\pm$ 0.137	0.158 $\pm$ 0.100	0.152 $\pm$ 0.092	0.132 $\pm$ 0.085	0.775 $\pm$ 0.155
	MiniLM	0.301 $\pm$ 0.172	0.131 $\pm$ 0.088	0.115 $\pm$ 0.070	0.099 $\pm$ 0.068	0.723 $\pm$ 0.138
	Mpnet	0.379 $\pm$ 0.150	0.163 $\pm$ 0.101	0.165 $\pm$ 0.095	0.138 $\pm$ 0.086	0.776 $\pm$ 0.157
Qwen3-14B	DRoberta	0.353 $\pm$ 0.135	0.139 $\pm$ 0.083	0.141 $\pm$ 0.082	0.114 $\pm$ 0.071	0.762 $\pm$ 0.144
	Pr-M-LM	0.335 $\pm$ 0.150	0.132 $\pm$ 0.091	0.130 $\pm$ 0.081	0.108 $\pm$ 0.073	0.762 $\pm$ 0.152
	Pr-M-Mp	0.360 $\pm$ 0.151	0.152 $\pm$ 0.094	0.151 $\pm$ 0.087	0.128 $\pm$ 0.080	0.780 $\pm$ 0.149
	MiniLM	0.322 $\pm$ 0.169	0.140 $\pm$ 0.092	0.131 $\pm$ 0.076	0.115 $\pm$ 0.074	0.736 $\pm$ 0.143
GPT-OSS-20B	Mpnet	0.380 $\pm$ 0.148	0.159 $\pm$ 0.093	0.173 $\pm$ 0.096	0.140 $\pm$ 0.083	0.780 $\pm$ 0.149
	DRoberta	0.356 $\pm$ 0.130	0.133 $\pm$ 0.082	0.144 $\pm$ 0.080	0.117 $\pm$ 0.069	0.765 $\pm$ 0.141
	Pr-M-LM	0.369 $\pm$ 0.131	0.152 $\pm$ 0.095	0.156 $\pm$ 0.091	0.130 $\pm$ 0.082	0.775 $\pm$ 0.150
	Pr-M-Mp	0.375 $\pm$ 0.163	0.164 $\pm$ 0.108	0.167 $\pm$ 0.104	0.141 $\pm$ 0.093	0.785 $\pm$ 0.157
Magistral-24B	MiniLM	0.315 $\pm$ 0.126	0.117 $\pm$ 0.076	0.111 $\pm$ 0.063	0.093 $\pm$ 0.059	0.721 $\pm$ 0.136
	Mpnet	0.297 $\pm$ 0.121	0.110 $\pm$ 0.068	0.102 $\pm$ 0.056	0.086 $\pm$ 0.055	0.715 $\pm$ 0.122
	DRoberta	0.275 $\pm$ 0.129	0.095 $\pm$ 0.062	0.084 $\pm$ 0.047	0.071 $\pm$ 0.045	0.704 $\pm$ 0.123
	Pr-M-LM	0.335 $\pm$ 0.125	0.126 $\pm$ 0.086	0.131 $\pm$ 0.081	0.105 $\pm$ 0.071	0.758 $\pm$ 0.152
Magistral-24B	Pr-M-Mp	0.345 $\pm$ 0.148	0.146 $\pm$ 0.094	0.147 $\pm$ 0.087	0.123 $\pm$ 0.079	0.771 $\pm$ 0.156
	MiniLM	0.311 $\pm$ 0.150	0.133 $\pm$ 0.084	0.107 $\pm$ 0.068	0.094 $\pm$ 0.064	0.715 $\pm$ 0.138
	Mpnet	0.279 $\pm$ 0.150	0.106 $\pm$ 0.081	0.092 $\pm$ 0.061	0.079 $\pm$ 0.060	0.689 $\pm$ 0.121
	DRoberta	0.253 $\pm$ 0.160	0.088 $\pm$ 0.062	0.080 $\pm$ 0.049	0.067 $\pm$ 0.047	0.683 $\pm$ 0.121
Magistral-24B	Pr-M-LM	0.327 $\pm$ 0.146	0.140 $\pm$ 0.091	0.133 $\pm$ 0.078	0.113 $\pm$ 0.073	0.759 $\pm$ 0.149
	Pr-M-Mp	0.346 $\pm$ 0.172	0.150 $\pm$ 0.097	0.147 $\pm$ 0.089	0.124 $\pm$ 0.081	0.773 $\pm$ 0.156

of reasoning, performance remains relatively limited, with accuracy values ranging from 26.3% to 34.4% and F1-scores below 13% across embedding models. This outcome reflects the intrinsic difficulty of the task, driven by extreme class imbalance, long-tailed subcategory distributions, and the limited semantic expressiveness of representations derived solely from comma-separated structured features.

When the proposed framework is applied, notable and systematic gains are observed across all evaluation metrics. Accuracy improvements are evident for most reasoning-embedding pairs, with particularly pronounced increases in recall and F1-score which are the metrics that are especially critical in multiclass complaint classification with rare categories. For example, compared to the strongest non-reasoning baseline (Paraphrase-Multilingual-Mpnet-Base-V2), which achieves an average F1-score of 12.5%, reasoning-enriched representations frequently surpass this value, reaching up to 14.1%. Similarly, AUC values consistently increase under the reasoning-based setting, indicating improved class separability and ranking performance even

when absolute classification accuracy remains modest. These results empirically validate that LLM-generated reasoning provides complementary semantic information that cannot be captured by surface-level feature concatenation alone.

A closer comparison among reasoning-capable LLMs reveals meaningful differences in their effectiveness. Models such as Qwen3-14B and Cogito-14B demonstrate stronger and more stable performance improvements across most embeddings, particularly in recall and F1-score, suggesting superior reasoning alignment with the complaint classification task. In contrast, GPT-OSS-20B and Magistral-24B exhibit more modest gains, with performance in some configurations approaching that of the non-reasoning baseline. This indicates that larger parameter counts do not necessarily translate to better reasoning utility in structured, domain-specific tasks, and that reasoning quality and instruction-following behavior are more critical than model scale alone. From the embedding perspective, Paraphrase-Multilingual-Mpnet-Base-V2 consistently yields the best average performance across both reasoning and non-reasoning settings. When paired with Qwen3-14B, it achieves the highest precision (16.4%), F1-score (14.1%), and AUC (78.5%) among all configurations. This superior performance can be attributed to its higher embedding dimensionality (768), stronger contextual encoding capacity, and multilingual robustness, which are particularly advantageous given the linguistic diversity and informal nature of municipal complaint text. The Paraphrase-Multilingual-MiniLM-L12-V2 model also performs competitively, offering a favorable trade-off between embedding compactness and semantic expressiveness.

Overall, the best-performing configuration, Qwen3-14B combined with Paraphrase-Multilingual-Mpnet-Base-V2 embeddings demonstrates that high-quality reasoning generation coupled with expressive embedding models yields the most discriminative representations. This combination effectively captures latent intent, contextual nuance, and inter-class distinctions, leading to consistent gains across all evaluation metrics. These findings underscore the central contribution of this work, showing that reasoning-enriched embeddings significantly enhance complaint classification performance, particularly in challenging real-world settings characterized by limited samples and highly imbalanced class distributions.

4.2 Average Performance Comparison of Majority Classes and Minority Complaint Subcategories

Unlike the previous subsection, which focused on overall average performance aggregated across all complaint categories, this subsection provides a frequency-aware performance analysis by explicitly separating classes according to their occurrence in the training data. The class frequency ranges reported in Tables 5 - 7 are derived using quantile-based partitioning of the empirical class frequency distribution, followed by rounding to ensure interpretable and discrete intervals. Specifically, complaint subcategories are grouped into four bins such as rare (< 27 instances), low-frequency (27 - 116), medium-frequency (117 - 476), and high-frequency (> 476) based on their position in the long-tailed distribution. This stratification allows a more detailed assessment of how the proposed reasoning-enriched framework behaves under varying degrees of data scarcity, which is critical in real-world complaint classification where most categories are underrepresented. By contrast with the previous subsection’s holistic

view, this analysis isolates the framework’s effectiveness for minority versus majority classes, offering deeper insights into generalization, effectiveness, and fairness across class frequencies.

Table 5 Average Performance (mean $\pm$ standard deviation) based on Class Frequency in Training Data

Class Frequency	Reasoning	Accuracy	Precision	Recall	F1-score
< 27	No	0.009 $\pm$ 0.085	0.009 $\pm$ 0.085	0.007 $\pm$ 0.080	0.008 $\pm$ 0.081
	DeepSeek-R1-14B	0.021 $\pm$ 0.120	0.021 $\pm$ 0.120	0.015 $\pm$ 0.102	0.017 $\pm$ 0.106
	Cogito-14B	0.019 $\pm$ 0.120	0.019 $\pm$ 0.120	0.015 $\pm$ 0.109	0.016 $\pm$ 0.112
	Qwen3-14B	0.022 $\pm$ 0.128	0.021 $\pm$ 0.126	0.017 $\pm$ 0.114	0.018 $\pm$ 0.117
	GPT-OSS-20B	0.012 $\pm$ 0.105	0.012 $\pm$ 0.105	0.011 $\pm$ 0.102	0.012 $\pm$ 0.102
	Magistral-24B	0.008 $\pm$ 0.079	0.008 $\pm$ 0.079	0.006 $\pm$ 0.073	0.007 $\pm$ 0.075
27 - 116	No	0.083 $\pm$ 0.198	0.074 $\pm$ 0.179	0.040 $\pm$ 0.143	0.048 $\pm$ 0.151
	DeepSeek-R1-14B	0.111 $\pm$ 0.236	0.101 $\pm$ 0.216	0.060 $\pm$ 0.185	0.070 $\pm$ 0.191
	Cogito-14B	0.115 $\pm$ 0.240	0.103 $\pm$ 0.220	0.063 $\pm$ 0.190	0.073 $\pm$ 0.196
	Qwen3-14B	0.130 $\pm$ 0.255	0.112 $\pm$ 0.227	0.070 $\pm$ 0.197	0.081 $\pm$ 0.204
	GPT-OSS-20B	0.085 $\pm$ 0.216	0.077 $\pm$ 0.200	0.048 $\pm$ 0.176	0.055 $\pm$ 0.181
	Magistral-24B	0.079 $\pm$ 0.198	0.071 $\pm$ 0.180	0.040 $\pm$ 0.152	0.047 $\pm$ 0.158
117 - 476	No	0.114 $\pm$ 0.204	0.068 $\pm$ 0.122	0.029 $\pm$ 0.087	0.037 $\pm$ 0.096
	DeepSeek-R1-14B	0.152 $\pm$ 0.240	0.087 $\pm$ 0.147	0.042 $\pm$ 0.117	0.052 $\pm$ 0.125
	Cogito-14B	0.151 $\pm$ 0.243	0.086 $\pm$ 0.150	0.043 $\pm$ 0.127	0.053 $\pm$ 0.133
	Qwen3-14B	0.182 $\pm$ 0.275	0.108 $\pm$ 0.188	0.063 $\pm$ 0.170	0.073 $\pm$ 0.174
	GPT-OSS-20B	0.120 $\pm$ 0.215	0.072 $\pm$ 0.139	0.034 $\pm$ 0.117	0.042 $\pm$ 0.122
	Magistral-24B	0.113 $\pm$ 0.203	0.068 $\pm$ 0.122	0.029 $\pm$ 0.089	0.037 $\pm$ 0.097
> 476	No	0.286 $\pm$ 0.309	0.089 $\pm$ 0.104	0.044 $\pm$ 0.084	0.053 $\pm$ 0.089
	DeepSeek-R1-14B	0.330 $\pm$ 0.325	0.100 $\pm$ 0.118	0.054 $\pm$ 0.103	0.064 $\pm$ 0.107
	Cogito-14B	0.336 $\pm$ 0.324	0.104 $\pm$ 0.121	0.057 $\pm$ 0.105	0.067 $\pm$ 0.109
	Qwen3-14B	0.347 $\pm$ 0.322	0.106 $\pm$ 0.122	0.059 $\pm$ 0.106	0.070 $\pm$ 0.110
	GPT-OSS-20B	0.287 $\pm$ 0.305	0.087 $\pm$ 0.092	0.040 $\pm$ 0.066	0.050 $\pm$ 0.072
	Magistral-24B	0.285 $\pm$ 0.306	0.090 $\pm$ 0.103	0.043 $\pm$ 0.083	0.052 $\pm$ 0.088

Table 5 presents the average performance across class-frequency groups without conditioning on a specific embedding model, thereby highlighting the pure impact of LLM-generated reasoning under varying data availability. For the rarest classes (< 27 instances), performance without reasoning is extremely limited, with accuracy, precision, recall, and F1-score all clustered around 0.8% - 0.9%. This confirms that conventional representations are largely incapable of learning meaningful decision boundaries when supervision is nearly absent. When reasoning is introduced, consistent improvements are observed across most LLMs. Qwen3-14B achieves the strongest results for rare classes, with an average accuracy of $2.2\% \pm 12.8\%$ and F1-score of $1.8\% \pm 11.7\%$, effectively more than doubling the baseline F1-score. DeepSeek-R1-14B and Cogito-14B also demonstrate noticeable gains, whereas Magistral-24B fails to improve over the baseline, indicating weaker reasoning utility in extreme few-shot scenarios.

In the low-frequency range (27 - 116 instances), the performance gap between reasoning and non-reasoning models becomes more pronounced. The baseline F1-score increases modestly to $4.8\% \pm 15.1\%$, reflecting the availability of limited supervision.

However, reasoning-capable models consistently outperform this baseline, with Qwen3-14B again leading at an F1-score of $8.1\% \pm 20.4\%$ and accuracy of $13\% \pm 25.5\%$. Cogito-14B and DeepSeek-R1-14B also provide substantial improvements, indicating that reasoning helps bridge the gap between sparse supervision and semantic generalization. GPT-OSS-20B and Magistral-24B, by contrast, yield only marginal gains, reinforcing the observation that reasoning quality, rather than model size alone, determines downstream effectiveness. For medium-frequency classes (117 - 476 instances), the baseline accuracy increases to $11.4\% \pm 20.4\%$ but recall and F1-score remain low at $2.9\% \pm 8.7\%$ and $3.7\% \pm 9.6\%$, respectively. Reasoning-based approaches significantly enhance performance such as Qwen3-14B nearly a twofold improvement over the baseline with an F1-score of $7.3\% \pm 17.4\%$. Similar trends are observed for Cogito-14B and DeepSeek-R1-14B. Finally, for majority classes (> 476 instances), absolute performance is substantially higher across all models due to enough class frequencies. Nonetheless, reasoning still yields consistent gains. For example, the baseline F1-score of $5.3\% \pm 8.9\%$ improves to $7\% \pm 11\%$ under Qwen3-14B.

Table 6 Average Performance (mean $\pm$ standard deviation) for Rare Classes (< 27 instances)

Reasoning	Embedding	Accuracy	Precision	Recall	F1-score
No	MiniLM	0.005 ± 0.071	0.005 ± 0.071	0.005 ± 0.071	0.005 ± 0.071
	Mpnet	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
	DRoberta	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
	Pr-M-LM	0.020 ± 0.132	0.020 ± 0.132	0.018 ± 0.125	0.019 ± 0.127
	Pr-M-Mp	0.017 ± 0.114	0.017 ± 0.114	0.013 ± 0.104	0.014 ± 0.106
DeepSeek-R1-14B	MiniLM	0.011 ± 0.088	0.011 ± 0.088	0.008 ± 0.076	0.009 ± 0.079
	Mpnet	0.018 ± 0.106	0.018 ± 0.106	0.011 ± 0.081	0.014 ± 0.088
	DRoberta	0.019 ± 0.118	0.019 ± 0.118	0.014 ± 0.105	0.016 ± 0.108
	Pr-M-LM	0.026 ± 0.141	0.026 ± 0.141	0.020 ± 0.127	0.022 ± 0.131
	Pr-M-Mp	0.030 ± 0.138	0.030 ± 0.138	0.020 ± 0.110	0.023 ± 0.118
Cogito-14B	MiniLM	0.010 ± 0.087	0.010 ± 0.087	0.008 ± 0.076	0.008 ± 0.079
	Mpnet	0.027 ± 0.151	0.027 ± 0.151	0.024 ± 0.144	0.025 ± 0.146
	DRoberta	0.008 ± 0.076	0.008 ± 0.079	0.006 ± 0.072	0.006 ± 0.073
	Pr-M-LM	0.027 ± 0.143	0.027 ± 0.143	0.021 ± 0.128	0.023 ± 0.131
	Pr-M-Mp	0.020 ± 0.122	0.020 ± 0.122	0.015 ± 0.106	0.017 ± 0.111
Qwen3-14B	MiniLM	0.012 ± 0.089	0.011 ± 0.088	0.008 ± 0.076	0.009 ± 0.079
	Mpnet	0.033 ± 0.162	0.032 ± 0.159	0.026 ± 0.146	0.028 ± 0.150
	DRoberta	0.022 ± 0.127	0.022 ± 0.123	0.016 ± 0.107	0.018 ± 0.112
	Pr-M-LM	0.020 ± 0.130	0.021 ± 0.131	0.017 ± 0.124	0.018 ± 0.126
	Pr-M-Mp	0.021 ± 0.122	0.021 ± 0.122	0.015 ± 0.105	0.017 ± 0.109
GPT-OSS-20B	MiniLM	0.012 ± 0.103	0.012 ± 0.103	0.011 ± 0.101	0.011 ± 0.101
	Mpnet	0.010 ± 0.101	0.010 ± 0.101	0.010 ± 0.101	0.010 ± 0.101
	DRoberta	0.010 ± 0.101	0.010 ± 0.101	0.010 ± 0.101	0.010 ± 0.101
	Pr-M-LM	0.013 ± 0.107	0.013 ± 0.107	0.011 ± 0.102	0.012 ± 0.103
	Pr-M-Mp	0.017 ± 0.114	0.017 ± 0.114	0.013 ± 0.104	0.014 ± 0.106
Magistral-24B	MiniLM	0.005 ± 0.071	0.005 ± 0.071	0.005 ± 0.071	0.005 ± 0.071
	Mpnet	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
	DRoberta	0.002 ± 0.024	0.002 ± 0.024	0.001 ± 0.008	0.001 ± 0.012
	Pr-M-LM	0.018 ± 0.117	0.018 ± 0.117	0.014 ± 0.105	0.015 ± 0.108
	Pr-M-Mp	0.014 ± 0.109	0.014 ± 0.109	0.012 ± 0.102	0.013 ± 0.104

Table 6 provides a deeper analysis of minority-class performance, explicitly examining the interaction between reasoning models and embedding architectures. Without reasoning, most embedding models completely fail to generalize to rare classes. For example, All-Mpnet-Base-V2 and All-DistilRoberta-V1 yield zero accuracy and F1-score, while All-MiniLM-L12-V2 achieves only $0.5\% \pm 7.1\%$ across all metrics. Even multilingual embeddings such as Paraphrase-Multilingual-MiniLM-L12-V2 and Paraphrase-Multilingual-Mpnet-Base-V2 exhibit limited effectiveness without reasoning, with F1-scores capped at $1.9\% \pm 12.7\%$ and $1.4\% \pm 10.6\%$, respectively. These results highlight the severe limitations of embedding-only approaches in extremely low-resource settings.

When reasoning is incorporated, performance improves systematically across nearly all embedding models. DeepSeek-R1-14B combined with Paraphrase-Multilingual-Mpnet-Base-V2 achieves an F1-score of $2.3\% \pm 11.8\%$, while Qwen3-14B with All-Mpnet-Base-V2 reaches $2.8\% \pm 15\%$, the highest F1-score observed for rare classes. These improvements, although numerically modest, are statistically meaningful given the near-zero baseline and the inherent difficulty of the task. Importantly, reasoning allows even monolingual embeddings such as All-Mpnet-Base-V2 to outperform multilingual baselines without reasoning, demonstrating that semantic reasoning can compensate for embedding limitations.

Across reasoning models, Qwen3-14B and DeepSeek-R1-14B consistently outperform GPT-OSS-20B and Magistral-24B, regardless of the embedding used. Magistral-24B exhibits minimal gains over the baseline, with several configurations remaining near zero performance. Overall, Table 6 underscores that the proposed framework is most impactful for minority classes, transforming an infeasible classification problem into one where meaningful, despite challenging, discrimination becomes possible.

Table 7 focuses on majority classes, where sufficient training data already enables moderate performance even without reasoning. In the baseline setting, accuracy ranges from $24.5\% \pm 29.3\%$ to $32.6\% \pm 31.9\%$, with the highest F1-score of $6\% \pm 8.6\%$ achieved by Paraphrase-Multilingual-Mpnet-Base-V2. These results indicate that embedding quality plays a more prominent role when data is abundant. However, recall remains relatively low across all baseline configurations, suggesting persistent confusion among semantically related high-frequency categories.

The introduction of reasoning consistently improves majority-class performance, particularly when paired with higher-capacity embeddings. The best overall results are obtained with Qwen3-14B and All-Mpnet-Base-V2, achieving an F1-score of $8.2\% \pm 13.2\%$, and Cogito-14B with All-DistilRoberta-V1, reaching $8.6\% \pm 16.3\%$. These gains demonstrate that reasoning enhances not only recall but also precision, indicating improved class boundary refinement rather than mere overprediction. Notably, reasoning benefits are more pronounced for embeddings that already capture rich contextual semantics, suggesting a complementary interaction between embedding expressiveness and reasoning quality. While the relative improvement for majority classes is smaller than for minority classes, the absolute gains remain meaningful in large-scale operational settings. Models such as GPT-OSS-20B and Magistral-24B again exhibit weaker improvements, reinforcing the broader trend observed throughout this study.

Table 7 Average Performance (mean $\pm$ standard deviation) for Majority Classes (> 476 instances)

Reasoning	Embedding	Accuracy	Precision	Recall	F1-score
No	MiniLM	0.290 $\pm$ 0.311	0.089 $\pm$ 0.097	0.043 $\pm$ 0.074	0.053 $\pm$ 0.081
	Mpnet	0.261 $\pm$ 0.304	0.089 $\pm$ 0.110	0.041 $\pm$ 0.091	0.049 $\pm$ 0.095
	DRoberta	0.245 $\pm$ 0.293	0.080 $\pm$ 0.108	0.037 $\pm$ 0.094	0.045 $\pm$ 0.097
	Pr-M-LM	0.309 $\pm$ 0.315	0.094 $\pm$ 0.105	0.047 $\pm$ 0.079	0.056 $\pm$ 0.085
	Pr-M-Mp	0.326 $\pm$ 0.319	0.094 $\pm$ 0.101	0.050 $\pm$ 0.080	0.060 $\pm$ 0.086
DeepSeek-R1-14B	MiniLM	0.304 $\pm$ 0.311	0.093 $\pm$ 0.108	0.046 $\pm$ 0.090	0.056 $\pm$ 0.094
	Mpnet	0.362 $\pm$ 0.338	0.120 $\pm$ 0.162	0.073 $\pm$ 0.154	0.084 $\pm$ 0.156
	DRoberta	0.321 $\pm$ 0.319	0.096 $\pm$ 0.110	0.048 $\pm$ 0.093	0.058 $\pm$ 0.096
	Pr-M-LM	0.314 $\pm$ 0.329	0.087 $\pm$ 0.097	0.047 $\pm$ 0.079	0.056 $\pm$ 0.084
	Pr-M-Mp	0.348 $\pm$ 0.327	0.103 $\pm$ 0.100	0.054 $\pm$ 0.077	0.065 $\pm$ 0.084
Cogito-14B	MiniLM	0.297 $\pm$ 0.313	0.088 $\pm$ 0.096	0.043 $\pm$ 0.072	0.052 $\pm$ 0.077
	Mpnet	0.368 $\pm$ 0.333	0.110 $\pm$ 0.106	0.060 $\pm$ 0.087	0.072 $\pm$ 0.093
	DRoberta	0.352 $\pm$ 0.329	0.126 $\pm$ 0.167	0.075 $\pm$ 0.161	0.086 $\pm$ 0.163
	Pr-M-LM	0.312 $\pm$ 0.323	0.093 $\pm$ 0.121	0.050 $\pm$ 0.100	0.060 $\pm$ 0.105
	Pr-M-Mp	0.351 $\pm$ 0.321	0.104 $\pm$ 0.100	0.055 $\pm$ 0.076	0.066 $\pm$ 0.082
Qwen3-14B	MiniLM	0.317 $\pm$ 0.314	0.093 $\pm$ 0.100	0.048 $\pm$ 0.075	0.058 $\pm$ 0.082
	Mpnet	0.372 $\pm$ 0.329	0.120 $\pm$ 0.140	0.070 $\pm$ 0.129	0.082 $\pm$ 0.132
	DRoberta	0.343 $\pm$ 0.311	0.101 $\pm$ 0.101	0.053 $\pm$ 0.078	0.064 $\pm$ 0.084
	Pr-M-LM	0.351 $\pm$ 0.325	0.110 $\pm$ 0.132	0.062 $\pm$ 0.121	0.073 $\pm$ 0.124
	Pr-M-Mp	0.350 $\pm$ 0.332	0.106 $\pm$ 0.129	0.063 $\pm$ 0.113	0.074 $\pm$ 0.118
GPT-OSS-20B	MiniLM	0.286 $\pm$ 0.307	0.080 $\pm$ 0.084	0.038 $\pm$ 0.061	0.048 $\pm$ 0.067
	Mpnet	0.269 $\pm$ 0.301	0.081 $\pm$ 0.078	0.033 $\pm$ 0.050	0.042 $\pm$ 0.056
	DRoberta	0.251 $\pm$ 0.293	0.085 $\pm$ 0.092	0.034 $\pm$ 0.060	0.043 $\pm$ 0.066
	Pr-M-LM	0.299 $\pm$ 0.312	0.092 $\pm$ 0.100	0.044 $\pm$ 0.075	0.054 $\pm$ 0.082
	Pr-M-Mp	0.328 $\pm$ 0.308	0.098 $\pm$ 0.104	0.051 $\pm$ 0.077	0.063 $\pm$ 0.085
Magistral-24B	MiniLM	0.288 $\pm$ 0.308	0.094 $\pm$ 0.096	0.041 $\pm$ 0.069	0.050 $\pm$ 0.074
	Mpnet	0.255 $\pm$ 0.302	0.080 $\pm$ 0.110	0.039 $\pm$ 0.091	0.048 $\pm$ 0.095
	DRoberta	0.240 $\pm$ 0.298	0.078 $\pm$ 0.107	0.037 $\pm$ 0.092	0.045 $\pm$ 0.095
	Pr-M-LM	0.305 $\pm$ 0.304	0.091 $\pm$ 0.089	0.043 $\pm$ 0.071	0.052 $\pm$ 0.075
	Pr-M-Mp	0.339 $\pm$ 0.311	0.108 $\pm$ 0.111	0.055 $\pm$ 0.088	0.067 $\pm$ 0.095

Collectively, the results in Table 7 confirm that the proposed reasoning-enriched framework is effective across the majority classes, delivering the largest relative benefits to minority classes while still providing consistent improvements for majority categories. This frequency-aware analysis further substantiates the framework’s suitability for real-world complaint classification systems characterized by extreme imbalance and evolving category distributions.

4.3 Ranking-Based Comparative Analysis of End-to-End Model Configurations

Table 8 presents a ranking-based comparison of end-to-end complaint classification models, showcasing the top 10 and bottom 10 configurations among all evaluated combinations of reasoning strategy, embedding model, and ML classifier. The ranking is determined by using the average score computed as the unweighted mean of the five-evaluation metrics namely accuracy, precision, recall, F1-score, and AUC to provide a

Table 8 Top 10 and Bottom 10 Ranked Model Configurations based on the Average Performance across Performance Metrics

Rank	Reasoning	Embedding Model		Accuracy	Precision	Recall	F1-score	AUC	Average
1	Qwen3-14B	Pr-M-Mp	LR	0.540	0.263	0.276	0.239	0.920	0.448
2	DeepSeek-R1-14B	Pr-M-Mp	LR	0.524	0.268	0.256	0.228	0.917	0.439
3	Cogito-14B	Mpnet	CatBoost	0.508	0.258	0.243	0.212	0.920	0.428
4	Qwen3-14B	Pr-M-LM	LR	0.511	0.232	0.262	0.215	0.914	0.427
4	Cogito-14B	Pr-M-Mp	LR	0.526	0.245	0.238	0.209	0.916	0.427
5	DeepSeek-R1-14B	Pr-M-LM	LR	0.501	0.250	0.248	0.214	0.914	0.425
5	Qwen3-14B	Mpnet	CatBoost	0.506	0.243	0.251	0.211	0.913	0.425
6	Magistral-24B	Pr-M-Mp	LR	0.513	0.235	0.250	0.209	0.912	0.424
7	No	Pr-M-Mp	LR	0.511	0.237	0.240	0.208	0.913	0.422
8	DeepSeek-R1-14B	Pr-M-Mp	CatBoost	0.484	0.253	0.236	0.216	0.906	0.419
8	Qwen3-14B	Pr-M-Mp	CatBoost	0.494	0.251	0.233	0.214	0.905	0.419
9	DeepSeek-R1-14B	Mpnet	CatBoost	0.505	0.242	0.236	0.204	0.905	0.418
10	GPT-OSS-20B	Pr-M-Mp	LR	0.499	0.227	0.237	0.201	0.912	0.415
139	GPT-OSS-20B	Pr-M-Mp	LGBM	0.088	0.002	0.013	0.003	0.502	0.122
140	DeepSeek-R1-14B	Mpnet	LGBM	0.077	0.003	0.016	0.003	0.504	0.121
140	No	MiniLM	LGBM	0.083	0.002	0.014	0.003	0.502	0.121
141	Qwen3-14B	Pr-M-Mp	LGBM	0.086	0.001	0.010	0.002	0.500	0.120
142	No	Mpnet	LGBM	0.076	0.009	0.009	0.002	0.500	0.119
143	Cogito-14B	Pr-M-LM	LGBM	0.078	0.001	0.009	0.002	0.500	0.118
144	Qwen3-14B	MiniLM	LGBM	0.063	0.002	0.015	0.003	0.503	0.117
144	Magistral-24B	Pr-M-Mp	LGBM	0.042	0.014	0.018	0.006	0.506	0.117
144	Magistral-24B	Mpnet	LGBM	0.076	0.001	0.008	0.002	0.499	0.117
145	No	Pr-M-LM	LGBM	0.035	0.012	0.010	0.002	0.500	0.112
146	No	Pr-M-Mp	LGBM	0.038	0.001	0.011	0.001	0.501	0.110
147	Cogito-14B	MiniLM	LGBM	0.020	0.010	0.010	0.001	0.500	0.108
148	Magistral-24B	DRoberta	LGBM	0.014	0.000	0.009	0.000	0.500	0.105

balanced assessment that does not favor any single metric. This analysis complements the previous performance comparisons by offering a holistic view of how complete pipelines behave in practice, highlighting not only the strongest-performing systems but also configurations that consistently fail to generalize.

The top-ranked models are dominated by configurations that integrate reasoning-capable LLMs, high-quality multilingual embeddings, and relatively simple yet effective classifiers. The best-performing configuration, Qwen3-14B with Paraphrase-Multilingual-Mpnet-Base-V2 and LR achieves the highest average score of 44.8%, with an accuracy of 54%, F1-score of 23.9%, and an AUC of 92%. This result indicates a strong balance between discrimination ability and class-wise prediction quality. The consistently high AUC values across the top-ranked models (all exceeding 91%) suggest that reasoning-enriched embeddings substantially improve the separability of complaint categories, even when downstream classifiers remain linear. Closely following are DeepSeek-R1-14B with Paraphrase-Multilingual-Mpnet-Base-V2 (LR) and Cogito-14B with All-Mpnet-Base-V2 (CatBoost), achieving average scores of 43.9% and 42.8%, respectively. These results reinforce the observation that reasoning quality and embedding expressiveness are more influential than classifier complexity. A

notable trend among the top configurations is the frequent appearance of LR and CatBoost, while more complex models such as LGBM are absent. LR appears in most top 10 entries, demonstrating that when embeddings are sufficiently informative and semantically structured through reasoning, linear decision boundaries are often sufficient. This finding is important from a practical perspective, as it suggests that the proposed framework not only improves performance but also enables the use of simpler, more interpretable, and computationally efficient classifiers. Additionally, the strong showing of Paraphrase-Multilingual-Mpnet-Base-V2 across top ranks highlights the value of multilingual contextual embeddings in handling heterogeneous complaint text which may contain informal language or domain-specific terminology.

Interestingly, one baseline configuration without reasoning in Paraphrase-Multilingual-Mpnet-Base-V2 and LR appears at rank 7th with an average score of 42.2%. While this indicates that high-quality embeddings alone can yield competitive performance, the fact that it is consistently outperformed by multiple reasoning-enabled counterparts underscores the incremental yet systematic benefit of LLM-generated reasoning. Even when absolute metric differences appear moderate, the consistent ranking advantage across multiple configurations demonstrates that reasoning provides an effective performance uplift rather than isolated gains. In stark contrast, the bottom-ranked models reveal clear and consistent failure patterns. All configurations ranked between 139 and 148 employ LGBM as the classifier, regardless of the reasoning model or embedding used. These models exhibit extremely low accuracy as low as 1.4%, near-zero F1-scores, and AUC values hovering around 50%, indicating behavior close to random guessing. For instance, the lowest-ranked configuration such as Magistral-24B with All-DistilRoberta-V1 and LGBM achieves an average score of only 10.5%, with an F1-score of 0% and an AUC of 50%. This suggests that LGBM struggles severely in this high-dimensional, extremely imbalanced multiclass setting, likely due to its sensitivity to sparse decision boundaries and limited effectiveness when class-level signal is weak. Another important observation is that reasoning alone is insufficient to guarantee strong performance if paired with an unsuitable classifier. Several reasoning-enabled configurations using Qwen3-14B, Cogito-14B, or DeepSeek-R1-14B still appear in the bottom ranks when combined with LGBM, achieving average scores below 12%. This highlights the importance of end-to-end compatibility where the benefits of reasoning-enriched embeddings can be nullified by downstream models that fail to exploit dense semantic representations effectively. Moreover, the near-identical AUC values across bottom-ranked models indicate that these systems are unable to learn meaningful ranking functions, regardless of reasoning or embedding quality.

To further complement the ranking-based evaluation, Table 9 presents an in-depth error analysis focusing on the two classes with the highest average error across all configurations, namely class 179 (*vehicle obstruction*) and class 122 (*other complaints related to tree*). These classes are particularly challenging due to their semantic overlap with other complaint categories and the presence of ambiguous or context-dependent textual descriptions. A clear distinction emerges between high-performing and low-performing configurations in terms of their error composition. The top-ranked models, particularly those combining reasoning-enabled LLMs (e.g., Qwen3-14B, Cogito-14B,

Table 9 Error Analysis of Misclassification Insights for Top 2 Error-Prone Subcategories (Class 179 = *vehicle obstruction*, Class 122 = *other complaints related to tree*) across Configurations

Rank	Reasoning	Embedding Model	Average Error	Class 179			Class 122			
				FPR	FNR	Error	FPR	FNR	Error	
1	Qwen3-14B	Pr-M-Mp	LR	0.005	0.033	0.121	0.042	0.050	0.083	0.053
2	Cogito-14B	Pr-M-Mp	LR	0.005	0.045	0.125	0.052	0.049	0.109	0.054
3	Deepseek-R1-14B	Pr-M-Mp	LR	0.005	0.040	0.133	0.049	0.049	0.105	0.054
146	No	Pr-M-LM	LGBM	0.011	0.008	0.984	0.102	0.000	0.991	0.086
147	Cogito-14B	MiniLM	LGBM	0.011	0.028	0.984	0.120	0.000	1.000	0.086
148	Magistral-24B	DRoberta	LGBM	0.011	0.002	1.000	0.098	0.000	1.000	0.086

and DeepSeek-R1-14B) with Paraphrase-Multilingual-Mpnet-Base-V2 embeddings and LR, achieve remarkably low average error (0.5%). Notably, these configurations maintain a balanced trade-off between FPR and FNR. For class 179, although the FNR remains relatively higher (ranging from 12.1% to 13.3%), the FPR is consistently controlled at low levels (3.3% - 4.5%), resulting in a modest overall error. A similar trend is observed for class 122, where both FPR and FNR remain comparatively low, with the best configuration achieving an error of 5.3%. This suggests that reasoning-enhanced embeddings contribute to improved semantic discrimination and help distinguish between subcategories in context.

In contrast, the bottom-ranked configurations reveal a fundamentally different failure mode, characterized by extreme imbalance between FPR and FNR. Across the low-performing models, particularly those using LGBM, the FNR approaches 100% for both classes, indicating that nearly all true instances are misclassified. For example, the Magistral-24B with All-DistilRoberta-V1 embedding and LGBM exhibits an FNR of 100% for both class 179 and 122, effectively failing to identify any true positives. Interestingly, this is accompanied by near-zero FPR (as low as 0%), suggesting that the model adopts an overly conservative prediction strategy, rarely assigning instances to these classes. While such behavior minimizes false alarms, it severely compromises recall, leading to high overall error (8.6% - 12%). This phenomenon reflects a well-known limitation in imbalanced multiclass settings, where certain models collapse to majority-class predictions when unable to learn discriminative boundaries.

Another important observation is that reasoning alone does not mitigate this issue when paired with an incompatible classifier. Even with advanced reasoning models such as Cogito-14B and Magistral-24B, the use of LGBM results in persistently high FNR values. This reinforces the earlier conclusion that the effectiveness of reasoning-augmented representations is highly dependent on the downstream model's ability to leverage dense semantic features. In this case, LGBM appears to underutilize the embedding space, leading to decision boundaries that fail to capture minority class signals. Without sufficient contextual reasoning, models may struggle to differentiate the distinctions between complaint subcategories, leading to either under-prediction (high FNR) or confusion with semantically adjacent classes.

Table 10 Statistical Significance Analysis Comparing Reasoning-based and Non-Reasoning Configurations

Statistical Test	Metric	Accuracy	Precision	Recall	F1-score	AUC
Shapiro-Wilk	Statistics	0.914	0.873	0.918	0.904	0.888
	p-value	0	0	0	0	0
	Reject H_0	Yes	Yes	Yes	Yes	Yes
Friedman	Statistics	195.002	184.655	182.855	188.299	194.262
	p-value	0	0	0	0	0
	Reject H_0	Yes	Yes	Yes	Yes	Yes

Table 11 Confidence Interval of Performance Metrics for Reasoning-based Frameworks versus the Baseline without Reasoning

Reasoning	Accuracy	Precision	Recall	F1-score	AUC
No	(0.302, 0.308)	(0.101, 0.113)	(0.076, 0.081)	(0.073, 0.078)	(0.648, 0.656)
Cogito-14B	(0.342, 0.349)	(0.156, 0.177)	(0.113, 0.119)	(0.109, 0.115)	(0.679, 0.688)
Deepseek-R1-14B	(0.344, 0.350)	(0.158, 0.180)	(0.111, 0.118)	(0.109, 0.116)	(0.675, 0.684)
GPT-OSS-20B	(0.310, 0.316)	(0.124, 0.138)	(0.091, 0.097)	(0.088, 0.094)	(0.655, 0.663)
Magistral-24B	(0.300, 0.306)	(0.127, 0.141)	(0.087, 0.092)	(0.086, 0.091)	(0.645, 0.653)
Qwen3-14B	(0.357, 0.363)	(0.167, 0.184)	(0.124, 0.132)	(0.120, 0.127)	(0.687, 0.696)

To rigorously assess the effectiveness of the proposed framework against the baseline without reasoning, statistical significance tests and confidence interval analyses are conducted, as presented in Tables 10 and 11. These analyses provide complementary evidence beyond point estimates, ensuring that the observed improvements are not due to random variation but reflect consistent and reliable performance gains. Table 10 first evaluates the distributional properties of the performance metrics using the Shapiro-Wilk test. The results indicate that all metrics significantly deviate from normality (p-value = 0), leading to the rejection of the null hypothesis for normal distribution. This justifies the use of non-parametric statistical tests for subsequent comparisons. Accordingly, the Friedman test is employed to assess whether there are statistically significant differences among the evaluated configurations. The Friedman statistics for all metrics are substantially large (ranging from 182.299 to 195.002), with p-values equal to 0, indicating strong evidence to reject the null hypothesis of equivalent performance. This confirms that the inclusion of reasoning approaches leads to statistically significant differences across all evaluation metrics when compared to the baseline without reasoning.

While statistical significance establishes the presence of differences, Table 11 further quantifies the magnitude and reliability of these improvements through confidence intervals. A clear and consistent trend is observed where all reasoning-enabled configurations outperform the baseline across every metric. The baseline without reasoning

achieves relatively low performance, with Accuracy confined to (30.2%, 30.8%), F1-score to (7.3%, 7.8%), and AUC to (64.8%, 65.6%). In contrast, reasoning-based models demonstrate substantial improvements, particularly in metrics that reflect class-wise prediction quality such as Precision, Recall, and F1-score.

Among the evaluated models, Qwen3-14B achieves the strongest performance across all metrics, with the highest confidence intervals for Accuracy (35.7%, 36.3%), F1-score (12%, 12.7%), and AUC (68.7%, 69.6%). Importantly, these intervals do not overlap with those of the baseline, indicating that the improvement is not only statistically significant but also practically meaningful. Similarly, Cogito-14B and DeepSeek-R1-14B exhibit competitive performance, with closely aligned confidence intervals that consistently exceed the baseline. These results reinforce the earlier observation that reasoning-enhanced embeddings improve the model’s ability to capture semantic nuances, leading to better discrimination across complaint subcategories.

Interestingly, not all reasoning models yield uniform gains. GPT-OSS-20B and Magistral-24B show only marginal improvements over the baseline, with overlapping or near-overlapping intervals in certain metrics such as Accuracy and AUC. This suggests that the effectiveness of the proposed framework is sensitive to the quality of reasoning generated by the underlying LLM. In other words, reasoning is beneficial, but its impact depends on how well the model can produce structured and contextually relevant explanations that enhance the embedding space.

In summary, the combined evidence from the ranking-based evaluation (Table 8), error analysis (Table 9), and statistical validation (Tables 10 - 11) demonstrates that the effectiveness of complaint classification systems is driven by the alignment between reasoning quality, embedding expressiveness, and classifier suitability. The top-ranked configurations consistently integrate reasoning-capable LLMs such as Qwen3-14B, DeepSeek-R1-14B, and Cogito-14B with Paraphrase-Multilingual-Mpnet-Base-V2 and simple classifiers such as LR, achieving superior overall performance and balanced class-wise behavior, particularly in reducing misclassification for challenging categories such as *vehicle obstruction* (class 179) and *other complaints related to tree* (class 122). In contrast, the error analysis reveals that poorly performing pipelines, especially those using LGBM, suffer from extreme false negative rates approaching 100%, indicating systematic failure to capture minority class signals despite the presence of reasoning. These observations are further reinforced by the statistical tests, where significant differences across all evaluation metrics confirm that reasoning-enhanced frameworks consistently outperform the baseline without reasoning, and by the non-overlapping confidence intervals which highlight the effectiveness and practical significance of these improvements. Collectively, the findings establish that reasoning is a critical enabler of performance gains, but its benefits are fully realized only when paired with semantically rich embeddings and compatible classifiers.

4.4 Ablation Study on the Contribution of Few-Shot Learning and Reasoning

To systematically quantify the contribution of each component in the proposed framework, an ablation study is conducted as shown in Table 12, focusing on the effects of few-shot learning and reasoning while maintaining a fixed high-quality embedding

Table 12 Ablation study Evaluating the Contribution of Few-Shot Learning and Reasoning, while Keeping the Embedding Model Fixed (Paraphrase-Multilingual-Mpnet-Base-V2)

Context	Reasoning	Model	Accuracy	Precision	Recall	F1-score	AUC
No	No	LR	0.511	0.237	0.240	0.208	0.913
		GNB	0.295	0.170	0.202	0.154	0.875
		RF	0.439	0.162	0.141	0.125	0.773
		AdaBoost	0.221	0.017	0.032	0.019	0.630
		XGBoost	0.416	0.193	0.179	0.170	0.818
		LGBM	0.038	0.001	0.011	0.001	0.501
		CatBoost	0.490	0.242	0.217	0.197	0.909
		Average	0.344	0.146	0.146	0.125	0.774
No	Qwen3-14B	LR	0.505	0.225	0.231	0.202	0.904
		GNB	0.287	0.154	0.183	0.139	0.856
		RF	0.426	0.168	0.128	0.112	0.778
		AdaBoost	0.213	0.017	0.030	0.019	0.630
		XGBoost	0.390	0.183	0.162	0.151	0.793
		LGBM	0.126	0.019	0.024	0.013	0.508
		CatBoost	0.458	0.216	0.200	0.181	0.895
		Average	0.344	0.140	0.137	0.117	0.766
Few-shot	Qwen3-14B	LR	0.540	0.263	0.276	0.239	0.920
		GNB	0.339	0.189	0.234	0.170	0.888
		RF	0.473	0.220	0.189	0.164	0.820
		AdaBoost	0.242	0.018	0.033	0.019	0.640
		XGBoost	0.450	0.204	0.197	0.182	0.823
		LGBM	0.086	0.001	0.010	0.002	0.500
		CatBoost	0.494	0.251	0.233	0.214	0.905
		Average	0.375	0.164	0.167	0.141	0.785

model, as it plays a fundamental role in converting unstructured inputs such as *Title*, *Details*, and *Address* into structured embedding representations. The baseline setting without contextual prompting and reasoning achieves an average accuracy of 34.4% and F1-score of 12.5%, establishing a reference point for subsequent comparisons. Despite the use of a multilingual embedding model, the performance remains limited, particularly in recall (14.6%) and F1-score (12.5%), indicating challenges in capturing minority classes and complex semantic relationships. Among the classifiers, LR and CatBoost exhibit relatively stronger performance, while LGBM performs poorly.

-Interestingly, introducing reasoning alone (Qwen3-14B without few-shot context) does not lead to performance improvement; instead, a slight degradation is observed across most metrics, with the average F1-score decreasing from 12.5% to 11.7% and AUC from 77.4% to 76.6%. This suggests that reasoning, when applied in isolation, may introduce noise or variability that is not effectively aligned with the downstream classification objective. One possible explanation is that zero-shot reasoning lacks sufficient task-specific grounding, leading to representations that are semantically rich but not optimally structured for discrimination. This highlights that reasoning quality alone is insufficient, and its effectiveness depends on how well it is guided toward the target task.

In contrast, the integration of few-shot learning with reasoning yields a substantial and consistent improvement across all evaluation metrics. The average accuracy increases to 37.5%, while the F1-score rises to 14.1% and AUC to 78.5%, representing the best performance among all configurations. Notably, the LR classifier achieves the highest individual performance with an accuracy of 54% and F1-score of 23.9%, surpassing both baseline and reasoning-only settings. Similar improvements are observed across GNB, RF, XGBoost, and CatBoost, indicating that the benefits of few-shot reasoning generalize across different model families. These results demonstrate that few-shot examples provide critical contextual guidance, enabling the reasoning model to generate more task-relevant and discriminative representations.

From a component-wise perspective, the findings reveal three key insights. First, high-quality embeddings are necessary but not sufficient, as evidenced by the modest baseline performance despite using a strong multilingual model. Second, reasoning must be properly aligned with the task through contextual grounding; otherwise, it may fail to translate into performance gains. Third, few-shot learning plays a pivotal role in unlocking the full potential of reasoning by anchoring it to representative examples, thereby enhancing both semantic coherence and class separability. Furthermore, the persistent underperformance of LGBM across all settings, including the few-shot configuration, underscores that improvements from reasoning and contextual prompting are contingent on the classifier’s ability to exploit dense semantic features. While most models benefit from the enhanced representations, LGBM remains unable to leverage these improvements effectively, resulting in near-random performance. This reinforces the importance of end-to-end compatibility within the framework. Overall, the ablation study provides strong empirical evidence that the proposed framework derives its effectiveness not from any single component, but from the synergistic interaction between reasoning, contextual prompting, and embedding representations.

Despite these improvements, several limitations should be acknowledged. The extreme class imbalance inherent in the dataset continues to pose a significant challenge, as many subcategories have insufficient samples (Table 2) for effective learning, even with reasoning enrichment. Moreover, the use of LLM-generated reasoning introduces potential bias, as the quality and consistency of reasoning outputs depend on the underlying LLM, prompt design, and K examples, which may propagate systematic errors or domain misinterpretations into the embeddings. The proposed leave-one-out few-shot strategy $N - 1$ incurs substantial computational overhead, with quadratic complexity due to repeated LLM inference for each instance, limiting scalability for large-scale or real-time applications. Thus, it is recommended to parallelize reasoning generation process across instances to mitigate the practical impact of the quadratic complexity. These factors highlight that, while the framework demonstrates clear empirical benefits, further optimization and careful consideration of deployment constraints are necessary for practical adoption.

5 Conclusion

Unlike prior studies that rely solely on structured features or direct end-to-end LLM inference, this study presented a reasoning-enriched representation learning framework

that enhances structured public complaint classification by incorporating LLM-generated reasoning embeddings under a few-shot learning paradigm. The findings demonstrate that transforming intermediate reasoning into semantic features consistently improves classification performance across multiple embedding models and classifiers, particularly for minority and low-frequency subcategories. Among all evaluated configurations, the combination of Qwen3-14B with Paraphrase-Multilingual-Mpnet-Base-V2 embeddings and LR emerged as the best-performing model, achieving the highest average score of 44.8%, with an accuracy of 54%, an F1-score of 23.9%, and an AUC of 92%, indicating strong ranking capability and class separability. Error analysis shows that it achieves low class-wise errors for class 179 (4.2%) and class 122 (5.3%), with reduced FPR and FNR indicating balanced control over both false alarms and missed detections. Statistical tests confirm significance, with the Shapiro-Wilk test rejecting normality ($p < 0.05$) and the Friedman test showing significant differences across models (statistics ≥ 182.855 , $p < 0.05$), supported by non-overlapping confidence intervals (F1-score improving from 7.3% - 7.8% to 12% - 12.7%). The ablation study further validates the framework, where few-shot reasoning improves accuracy from 34.4% to 37.5% and F1-score from 12.5% to 14.1%, confirming its critical role in performance gains. These results highlight a key strength of the proposed approach: its ability to bridge reasoning-capable LLMs with existing ML pipelines, enabling semantically enriched representations that improve learning even in highly imbalanced and complex multiclass settings.

-However, several limitations and underlying assumptions should be considered when applying the results. The extreme class imbalance in the dataset presents a significant challenge, leading to relatively modest F1-scores for both the baseline and the proposed framework. In addition, the dependence on LLM-generated reasoning introduces potential bias and variability due to various behaviour, prompt formulation, and the selected K examples. The framework also assumes access to Graphics Processing Unit (GPU) resources, as the repeated LLM inference and embedding generation processes are computationally intensive and impractical to execute efficiently on standard Central Processing Unit (CPU)-based systems.

From a practical perspective, the proposed framework is best suited as a decision-support tool for municipal complaint management systems rather than a standalone solution. Its strong AUC performance indicates that it can effectively prioritize and rank complaints, assisting human operators in triaging cases to improve response efficiency. Looking forward, several directions are recommended to improve both applicability and impact. Future work should focus on addressing class imbalance through advanced techniques such as data augmentation, while also exploring methods to mitigate LLM bias and improve reasoning consistency. Reducing computational complexity through strategies such as reasoning caching, and batch inference will be critical for real-world scalability. Additionally, evaluating the framework across multilingual and cross-domain datasets with optimal K value will further validate its generalizability. Overall, this study provides a foundation for integrating LLM-based reasoning into structured data pipelines, while highlighting important considerations for practical deployment and future research.

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Declaration of interest

The authors declare no competing interest.

Consent to Publish Declaration

Not applicable

Consent to Participate Declaration

Not applicable

Ethics Declaration

Not applicable

Data availability statements

This work is not a clinical trial, thereby there are no registry, trial registration number, and data of registration provided. The data supporting the findings of this study are not publicly available due to their sensitive nature and the potential risk of compromising data privacy. The data are, however, available from the corresponding author upon reasonable request.

Author contributions

Theng-Jia Law: Conceptualization, Methodology, Validation, Formal analysis, Writing - Original Draft, Writing - Review & Editing, Visualization. **Choo-Yee Ting:** Supervision, Project administration, Writing - Review & Editing.

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